Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The radius of a circle is uniformly increasing at the rate of $3\ \mathrm{cm/s}$. What is the rate of increase in area, when the radius is $10\ \mathrm{cm}$?

(A) $6\pi \mathrm{cm}^2/\mathrm{s}$

(B) $10\pi \mathrm{cm}^2/\mathrm{s}$

(C) $30\pi \mathrm{cm}^2/\mathrm{s}$

(D) $60\pi \mathrm{cm}^2/\mathrm{s}$

Q.2

If water is poured into an inverted hollow cone whose semi-vertical angle is 30°. Its depth (measured along the axis) increases at the rate of $1\mathrm{cm/s}$. The rate at which the volume of water increases when the depth is $24\ \mathrm{cm}$ is

(A) $162\ \mathrm{cm}^3/\mathrm{s}$

(B) $172\ \mathrm{cm}^3/\mathrm{s}$

(C) $182\ \mathrm{cm}^3/\mathrm{s}$

(D) $192\ \mathrm{cm}^3/\mathrm{s}$

Q.3

If at any instant t, for a sphere, r denotes the radius, S denotes the surface area and V denotes the volume, then what is $\frac{dV}{dt}$ equal to?

(A) $\frac{1}{2}S\frac{dr}{dt}$

(B) $\frac{1}{2}r\frac{dS}{dt}$

(C) $r\frac{dS}{dt}$

(D) $\frac{1}{2}r^2\frac{dS}{dt}$

Q.4

The velocity v of a particle at any instant t moving in a straight line is given by $v = s + 1$ where 's' is in metre which is the place of the circle distance travelled in t second. What is the time taken by the particle to cover a distance of $9\ \mathrm{m}$?

(A) $1\ \mathrm{s}$

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(B) $(\log 10)s$

(C) $2(\log 10)s$

(D) 10 s

Q.5

A lamp is 50ft. above the ground. A ball is dropped from the same height from a point 30ft. away from the light pole. If ball falls a distance $s = 16t^2\text{ft}$. in t seconds, then the speed of the shadow of the ball moving along the ground $1/2$ s later is

(A) -1500ft/s

(B) 1500ft/s

(C) -1600ft/s

(D) 1600ft/s

Q.6

The approximate value of $(0.007)^{1/3}$

(A) $\frac{23}{120}$

(B) $\frac{27}{120}$

(C) $\frac{19}{120}$

(D) $\frac{17}{120}$

Q.7

An equilateral triangle's sides increase at the rate of 2 cm / sec. If the area of its incircle increases at a rate of k cm^2 / sec (when the length of the side is $\frac{6}{\pi}$cm), then the value of k is

(A) 2

(B) 3

(C) 5

(D) 7

Q.8

A man is moving away from a tower 41.6 m high at a rate of 2 m s^{-1}.. If the eye level of the man is 1.6 m above the ground, then the rate at which the angle of elevation of the top of the tower changes, when he is at a distance of 30 m from the foot of the tower, is

(A) $-\frac{4}{125}$ radian/s

(B) $-\frac{2}{25}$ radian/s

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(C) $-\frac{1}{625}$ radian/s

(D) None of these

Q.9

The rate of increase of bacteria in a certain culture is proportional to the number present. If it doubles in 5 hours then in 25 hours, its number would be

(A) 8 times the original

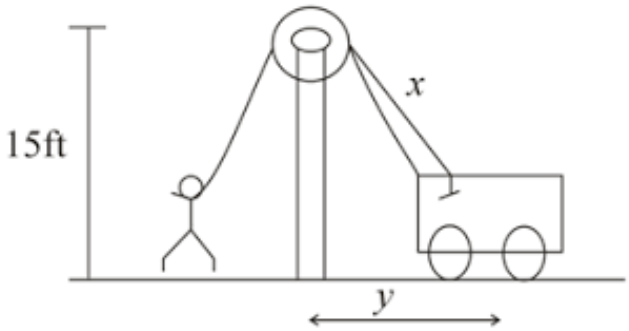
(B) 16 times the original

(C) 32 times the original

(D) 64 times the original

Q.10

A rope is attached to a pulley mounted on a $15 \quad ft$ tower. The end of the rope is attached to a heavily loaded cart (as shown in figure).



A worker can pull the rope at a rate of $2 \quad ft / s$. The cart is approaching the tower when it is $8 \quad ft$ from the tower at the rate of:

(A) $\frac{17}{4}$ ft/s

(B) $\frac{9}{4}$ ft/s

(C) $\frac{13}{4}$ ft/s

(D) None of these

Q.11

A lamp of negligible height is placed on the ground l_1 away from a wall. A man l_2 m tall is walking at a speed of $\frac{l_1}{10}$ m/s from the lamp to the nearest point on the wall. When he is midway between the lamp and the wall, the rate of change in the length of this shadow on the wall is

(A) $-\frac{5l_2}{2}$ m/s

(B) $-\frac{2l_2}{5}$ m/s

(C) $-\frac{l_2}{2}$ m/s

(D) $-\frac{l_2}{5}$ m/s

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Q.12

If a circular plate is heated uniformly, its area expands $3c$ times as fast as its radius, then the value of c when the radius is 6 units, is

(A) 4π

(B) 2π

(C) 6π

(D) 3π

Q.13

Two cyclists start from the junction of two perpendicular roads, their velocities being $3v\ \mathrm{m}/$ minute and $4v\ \mathrm{m}/$ minute. The rate at which the two cyclists are separating is

(A) $\frac{7}{2}v\ \mathrm{m}/$ minute

(B) $5\mathrm{vm}/\mathrm{minute}$

(C) $v\ \mathrm{m}/\mathrm{minute}$

(D) None of these

Q.14

A ball is dropped from a platform $19.6\ \mathrm{m}$ high. Its position function is -

(A) $x = -4.9t^2 + 19.6(0 \leq t \leq 1)$

(B) $x = -4.9t^2 + 19.6(0 \leq t \leq 2)$

(C) $x = -9.8t^2 + 19.6(0 \leq t \leq 2)$

(D) $x = -4.9t^2 - 19.6(0 \leq t \leq 2)$

Q.15

A particle moves along the curve $6\mathrm{y} = \mathrm{x}^3 + 2.$ The point ' P' on the curve at which the y -coordinate is changing 8 times as fast as the x -coordinate, which of the following are correct about the point 'P'...

(A) x -coordinates at the point P are ± 4

(B) y -coordinates at the point P are 11 and $\frac{-31}{3}$

(C) Both (a) and (b)

(D) None of the above

Q.16

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The diagonal of a square is changing at the rate of 0.5 cm/ sec. Then the rate of change of area, when the area is 400 cm^2, is equal to

(A) $20\sqrt{2} \text{ cm}^2/\text{sec}$

(B) $10\sqrt{2} \text{ cm}^2/\text{sec}$

(C) $\frac{1}{10\sqrt{2}} \text{ cm}^2/\text{sec}$

(D) $\frac{10}{\sqrt{2}} \text{ cm}^2/\text{sec}$

Q.17

A man of height 1.8 metre is moving away from a lamp post at the rate of 1.2 m/sec. If the height of the lamp post be 4.5 metre, then the rate at which the shadow of the man is lengthening is

(A) 0.4 m/sec

(B) 0.8 m/sec

(C) 1.2 m/sec

(D) None of these

Q.18

If a spherical balloon has a variable diameter $3x + \frac{9}{2}$, then the rate of change of its volume with respect to x is

(A) $27\pi(2x + 3)^2$

(B) $\frac{27\pi}{16}(2x + 3)^2$

(C) $\frac{27\pi}{8}(2x + 3)^2$

(D) None of these

Q.19

A spherical ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is

(A) $\frac{1}{54\pi}\text{cmmin}$

(B) $\frac{5}{6\pi}\text{cm/min}$

(C) $\frac{5}{36\pi}\text{cmmin}$

(D) $\frac{1}{18\pi}cm/\text{ min}$

Q.20

If $x < 1$, then the value of $1 - 7x^{\frac{1}{3}}1 + 2x^{-\frac{3}{4}}$ is ____

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(A) $1+\frac{23}{6}x$

(B) $1-\frac{23}{6}x$

(C) $1-\frac{25}{6}x$

(D) $1+\frac{25}{6}x$

Answers & Solutions

Q.1 Answer:

$60\pi \mathrm{cm}^2/\mathrm{s}$

Solution:

Given $\frac{\mathrm{dr}}{\mathrm{dt}}=3$

Let $\mathrm{A}=$ **Area of circle** $=\pi \mathrm{r}^2$

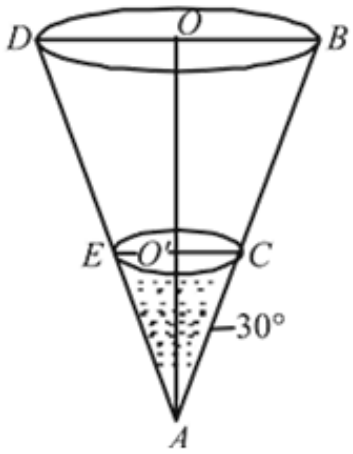
$\therefore \frac{\mathrm{dA}}{\mathrm{dt}}=2\pi \mathrm{r}\cdot\frac{\mathrm{dr}}{\mathrm{dt}}=6\pi \mathrm{r}$

Now, $\frac{\mathrm{dA}}{\mathrm{dt}}\Big|_{\mathrm{r}=10}=6\times 10\times\pi=60\pi \mathrm{cm}^2/\mathrm{s}$

Q.2 Answer:

$192\ \mathrm{cm}^3/\mathrm{s}$

Solution:



Let A **be the vertex and** AO **the axis of the cone. Let** $O'A=h$ **be the depth of water in the cone. In** $\triangle AO'C$,

$\tan 30^\circ=\frac{O'C}{h}$ **or** $O'C=\frac{h}{\sqrt{3}}=$ **radius**

$V=$ **Volume of water in the cone** $=\frac{1}{3}\pi\left(O'C\right)^2\times AO'$

$=\frac{1}{3}\pi\left(\frac{h^2}{3}\right)\times h=\frac{\pi}{9}h^3$

or $\frac{dV}{dt}=\frac{\pi}{3}h^2\frac{dh}{dt}$ **...(1)**

But given that depth of water increases at the rate of $1\ \mathrm{cm/s}$**. So,** $\frac{dh}{dt}=1\ \mathrm{cm/s}$ **...(2)**

From (1) **and** (2), $\frac{dV}{dt}=\frac{\pi h^2}{3}$

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When $h = 24$ cm, the rate of increase of volume is

$\frac{dV}{dt} = \frac{\pi(24)^2}{3} = 192 \text{ cm}^3/\text{s}$

Q.3 Answer:

$\frac{1}{2} r \frac{dS}{dt}$

Solution:

Surface area of sphere $S = 4\pi r^2$ **Differentiate both sides w.r.t. 't'** $\Rightarrow \quad \frac{dS}{dt} = \frac{8\pi r dr}{dt}$

and Volume $= V = \frac{4}{3}\pi r^3$

$\Rightarrow \quad \frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$

$= \frac{4\pi r^2}{8\pi r} \cdot \frac{dS}{dt} = \frac{1}{2} r \frac{dS}{dt}$

Q.4 Answer:

$(\log 10)s$

Solution:

Given velocity is $v = s + 1$ **Since, velocity** $= \frac{ds}{dt}$

$\therefore \frac{ds}{dt} = s + 1 \Rightarrow \frac{ds}{s+1} = dt$

Integrate both side we get $\log(s + 1) = t$

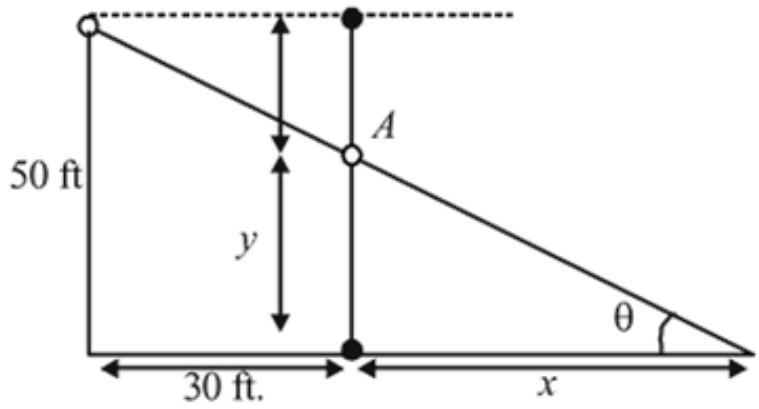
At $s = 9$ m

$t = \log(10)$ **second**

Q.5 Answer:

-1500ft/s

Solution:



At time t**, ball drops** $16t^2$ft. **distance.**

Therefore, $y = 50 - 16t^2$ **...(1)**

Point A **is the position of the falling ball at some time** t**.**

So, $\frac{dy}{dt} = -32t$ **...(2)**

From the figure, $\tan\theta = \frac{y}{x} = \frac{50}{30+x}$ **or**

$y = \left(\frac{50}{30+x}\right) \cdot x$

$\therefore \frac{dy}{dt} = \frac{d}{dt}\left(\frac{50x}{30+x}\right) = \frac{1500}{(30+x)^2} \cdot \frac{dx}{dt}$

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or $\frac{dx}{dt} = \frac{(30+x)^2}{1500}(-32t)$

When $t = \frac{1}{2}, y = 46 \quad$ [**using** (1)]

and $x = 345 \quad$ [**using** (2)]

$\therefore \frac{dx}{dt} = -16\frac{(375)^2}{1500} = -1500\text{ft/s}$

Q.6 Answer:

$\frac{23}{120}$

Solution:

Let $f(x) = x^{1/3} \Rightarrow \quad f'(x) = \frac{1}{3}x^{-2/3}$

Now $f(x+\Delta x) - f(x) = f'(x)\cdot\Delta x = \frac{\Delta x}{3\left(x^{2/3}\right)}$

We may write, $0.007 = 0.008 - 0.001$, **taking.** $x = 0.008$ **and** $dx = -0.001$.

we have $f(0.007) - f(0.008) = -\frac{0.001}{3(0.008)^{2/3}}$

$\Rightarrow f(0.007) - (0.008)^{1/3} = -\frac{0.001}{3(0.2)^2}$

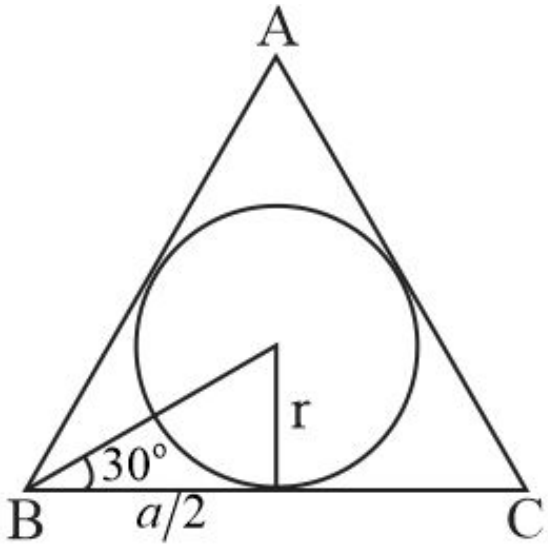
$\Rightarrow f(0.007) = 0.2 - \frac{0.001}{3(0.04)} = 0.2 - \frac{1}{120} = \frac{23}{120}$

Hence $(0.007)^{1/3} = \frac{23}{120}$

Q.7 Answer:

2

Solution:



$\frac{r}{\frac{a}{2}} = \tan 30^\circ$

$\frac{2r}{a} = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad r = \frac{a}{2\sqrt{3}}$

$\therefore$ **Area of incircle,** $A = \pi r^2 = \frac{\pi a^2}{12}$

$\therefore \quad \frac{dA}{dt} = \frac{2\pi a}{12}\frac{da}{dt}$

$= \frac{\pi}{6}\cdot\frac{6}{\pi}\cdot 2$

$= 2\text{cm}^2 / \text{sec}.$

Q.8 Answer:

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$-\frac{4}{125}$ radian/s

Solution:

Let CD be the position of man at any time t. Let $BD = x$, then $EC = x$. Let $\angle ACE = \theta$ Given,

$AB = 41.6 \quad m, CD = 1.6 \quad m$ **and** $\frac{dx}{dt} = 2 \quad \text{m} \quad \text{s}^{-1}$

$AE = AB - EB = AB - CD = 41.6 - 1.6 = 40 \quad m$

We have to find $\frac{d\theta}{dt}$ **when** $x = 30 \quad m$

From $\Delta AEC, \tan\theta = \frac{AE}{EC} = \frac{40}{\text{x}} \ldots 1$

Differentiating w.r.t. to $t, \sec^2\theta \frac{d\theta}{dt} = \frac{-40}{x^2}\frac{dx}{dt}$

$\Rightarrow sec^2\theta \frac{d\theta}{dt} = \frac{-40}{x^2} \times 2$

$\Rightarrow \frac{d\theta}{dt} = \frac{-80}{x^2} cos^2\theta = -\frac{80}{x^2}\frac{x^2}{x^2 + 40^2} \quad \because cos\theta = \frac{x}{\sqrt{x^2 + 40^2}}$

$\Rightarrow \frac{d\theta}{dt} = -\frac{80}{x^2 + 40^2}$

when $x = 30, \frac{d\theta}{dt} = -\frac{80}{30^2 + 40^2} = -\frac{4}{125}$ radian s^{-1}

Q.9 Answer:

32 times the original

Solution:

Let N **be the no. of bacteria at time t.**

Let N_0 **be the initial original no. of bacteria.**

Then, $\frac{\text{d}}{\text{dt}}\text{N} \propto \text{N} \frac{\text{d}}{\text{dt}}\text{N} = \text{kN} \Rightarrow \frac{\text{dN}}{\text{N}} = \text{kdt}$

$\Rightarrow \quad \int \frac{\text{dN}}{\text{N}} = \text{k} \int \text{dt} \Rightarrow \log \text{N} = \text{kt} + \text{c}$

At $\text{t} = 0, \ \text{N} = \text{N}_0 \Rightarrow \log \text{N}_0 = 0 + \text{c} \Rightarrow \text{c} = \log \text{N}_0$

$\therefore \log N = \text{kt} + \log \text{N}_0 \Rightarrow \log \frac{\text{N}}{\text{N}_0} = \text{kt}$

When $\text{t} = 5\text{hrs}.$

$\text{N} = 2\,\text{N}_0$

$\therefore \log \left(\frac{2\,\text{N}_0}{\text{N}_0}\right) = 5\text{k}$

$\Rightarrow \text{k} = \frac{\log 2}{5}$

$\therefore \log \frac{\text{N}}{\text{N}_0} = \frac{\log 2}{5}\text{t}$

When $\text{t} = 25\text{hrs}, \log \frac{\text{N}}{\text{N}_0} = \frac{\log 2}{5} \times 25$

$\Rightarrow \log \frac{\text{N}}{\text{N}_0} = 5 \log 2$

$\log \frac{\text{N}}{\text{N}_0} = \log 2^5$

$\Rightarrow \frac{\text{N}}{\text{N}_0} = 32 \quad \therefore \text{N} = 32\,\text{N}_0$

Therefore, 32 times the original.

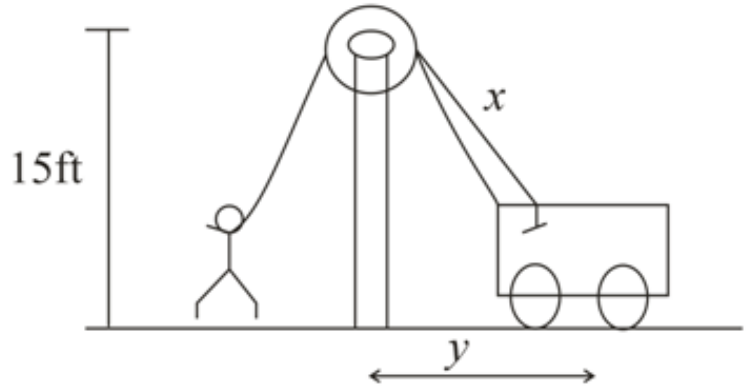
Q.10 Answer:

$\frac{17}{4}$ ft/s

Solution:

Let y **be the distance of the cart from the tower at any time** t**.**

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So, $x^2 = \quad y^2 + 15^2$

Given, $\frac{dx}{dt} = 2 \quad ft/s$

So, differentiating both sides of $x^2 = y^2 + 15^2$ **with respect to time** t**, we get:**

$\Rightarrow 2x\frac{dx}{dt} \quad = \quad 2y\frac{dy}{dt}$

$\Rightarrow x\frac{dx}{dt} \quad = \quad y\frac{dy}{dt}$

It is given that, $y = 8 \quad ft$

So, $x^2 = 8^2 + 15^2$

$= 64 + 225$

$= 289$

$\Rightarrow x = 17 \quad ft$

Also, $\frac{dx}{dt} = 2 \quad ft/s$

Therefore, $x\frac{dx}{dt} \quad = \quad y\frac{dy}{dt}$

$\Rightarrow \frac{dy}{dt} = \frac{x}{y} \quad \frac{dx}{dt}$

$= \frac{17}{8} \times 2 \quad = \quad \frac{17}{4} \quad ft/s$

Therefore, the cart is approaching the tower when it is $8 \quad ft$ **from the tower at the rate of** $\frac{17}{4} \quad ft/s$**.**

Q.11 Answer:

$-\frac{2l_2}{5}\ \mathrm{m/s}$

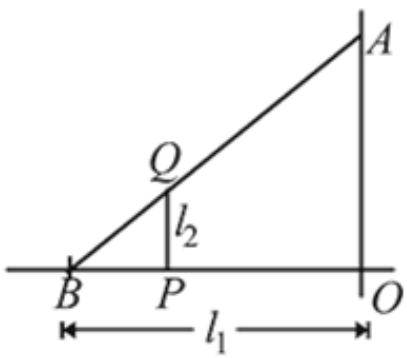
Solution:

Let $BP = x$**. From similar triangle property, we get**

$\frac{AO}{l_1} = \frac{l_2}{x}$ **or** $AO = \frac{l_1 l_2}{x}$

or $\frac{d(AO)}{dt} = \frac{-l_1 l_2}{x^2}\frac{dx}{dt}$

When, $x = \frac{l_1}{2}, \frac{d(AO)}{dt} = -\frac{2l_2}{5}\ \mathrm{m/s}$



Q.12 Answer:

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4π

Solution:

Let A sq. units in the area measure when the radius is r units. their $A = \pi r^2$
Differentiate both side w.r.t ' t ' $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$ ...(i)
We have, $\frac{dA}{dt} = 3c\frac{dr}{dt}$
Fromeqn (i), we get 3c. $\frac{dr}{dt} = 2\pi r \cdot \frac{dr}{dt} \Rightarrow 3c = 2\pi$
Now, $c = \frac{2}{3}\pi(6) = 4\pi$ when $r = 6$

Q.13 Answer:

$5\mathrm{vm/minute}$

Solution:

At time t, the distance z between the
cyclists is given by $z^2 = (3vt)^2 + (4vt)^2$
$\therefore z = 5vt \Rightarrow \frac{dz}{dt} = 5v$

Q.14 Answer:

$x = -4.9t^2 + 19.6(0 \leq t \leq 2)$

Solution:

We have, $\mathrm{a} = \frac{\mathrm{d}^2\mathrm{x}}{\mathrm{dt}^2} = -9.8$
The initial conditions are $x(0) = 19.6$ and $v(0) = 0$
So, $\mathrm{v} = \frac{\mathrm{dx}}{\mathrm{dt}} = -9.8\mathrm{t} + \mathrm{v}(0) = -9.8\mathrm{t}$
$x = -4.9t^2 + x(0) = -4.9t^2 + 19.6$
Now, the domain of the function is restricted since the ball hits the ground after a certain time. To find this time we set $x = 0$ and solve for t :
$0 = -4.9\mathrm{t}^2 + 19.6 \Rightarrow \mathrm{t} = 2$

Q.15 Answer:

Both (a) and (b)

Solution:

Given, $6y = x^3 + 2$
On differentiating w.r.t. t, we get $6\frac{\mathrm{dy}}{\mathrm{dt}} = 3\mathrm{x}^2\frac{\mathrm{dx}}{\mathrm{dt}} \quad \Rightarrow \quad 6 \times 8\frac{\mathrm{dx}}{\mathrm{dt}} = 3\mathrm{x}^2\frac{\mathrm{dx}}{\mathrm{dt}}$
$\Rightarrow \quad 3x^2 = 48 \quad \Rightarrow \quad x^2 = 16$
$\Rightarrow \quad x = \pm 4$
When $x = 4$, then $6y = (4)^3 + 2$
$\Rightarrow \quad 6y = 64 + 2 \Rightarrow y = \frac{66}{6} = 11$
When $x = -4$, then $6y = (-4)^3 + 2$
$\Rightarrow \quad 6y = -64 + 2 \Rightarrow y = \frac{-62}{6} = \frac{-31}{3}$
Hence, the required points on the curve are $(4, 11)$ and $\left(-4, \frac{-31}{3}\right)$

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Q.16 Answer:

$10\sqrt{2}\ \text{cm}^2/\text{sec}$

Solution:

Diagonal $\text{D} = \sqrt{2}.\text{a}$

Differentiating w.r.t. t

$\frac{\text{dD}}{\text{dt}} = \sqrt{2}\frac{\text{da}}{\text{at}}$ **or** $\frac{\text{da}}{\text{dt}} = \frac{1}{\sqrt{2}}\frac{\text{da}}{\text{dt}} = \frac{1}{\sqrt{2}} \times 0.5\ \text{cm/s}$

Let Area is denoted by A

$\frac{\text{dA}}{\text{dt}} = 2\text{a}\frac{\text{da}}{\text{dt}}$ **... (i)**

when area A **is** $400\ \text{cm}^2$ **then** $\text{a} = 20$

$\therefore \frac{\text{dA}}{\text{dt}} = 2 \times 20 \times \frac{0.5}{\sqrt{2}} = 10\sqrt{2}\ \text{cm}^2/\text{sec}$

Q.17 Answer:

$0.8\ \text{m/sec}$

Solution:

$\frac{dy}{dt} = 1.2$ **According to the figure**

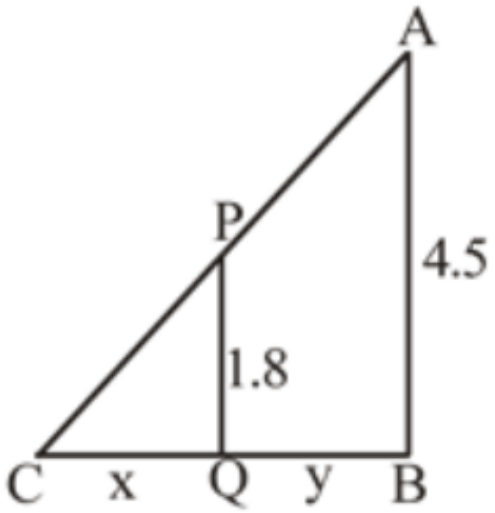
$\frac{4.5}{1.8} = \frac{x+y}{x}$

$\frac{5}{2} = \frac{x+y}{x} \Rightarrow 3x = 2y$

$x = \frac{2}{3}y \Rightarrow \frac{dx}{dt} = \frac{2}{3} \cdot \frac{dy}{dt}$

∴ Required rate of length of shadow

$\frac{dx}{dt} = 0.8\ \text{m/s}$



Q.18 Answer:

$\frac{27\pi}{8}(2x+3)^2$

Solution:

Radius of balloon

$= r = \frac{3}{4}(2x+3) \Rightarrow \frac{d}{dx} = \frac{3}{2}$

∴ Rate of change in volume $= 4\pi\left(\frac{3}{4}\right)^2(2x+3)^2 \cdot \frac{3}{2} = \frac{27\pi}{8}(2x+3)^2$

Q.19 Answer:

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$\frac{1}{18\pi} cm/\min$

Solution:

$\frac{dv}{dt} = 50 \text{ cm}^3/\text{min}$

let radius of ball $= rcm$

$\therefore$ **volume** $v = \frac{4}{3}\pi^3$

$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$

when $r = 10 + 5 = 15$

$50 = 4\pi \times (15)^2 \frac{dr}{dt}$

$\frac{50}{4\pi \times 225} = \frac{dr}{dt}$

$\therefore \frac{dr}{dt} = \frac{1}{18\pi} \text{cm/min}$

Q.20 Answer:

$1 - \frac{23}{6} \quad x$

Solution:

$1 - 7x^{\frac{1}{3}} \cdot 1 + 2x^{-\frac{3}{4}}$

$= 1 - \frac{7}{3}x1 - \frac{3}{2}x$ **nearly**

$= 1 - \frac{7}{3}x - \frac{3}{2} \times$ **nearly**

$= 1 - \frac{23}{6} \quad x$

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